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Inputs

Area	Variable	Value	Units	Reference	Rev
	Intake air properties				
	Ambient temperature	21	°C		A
	Idea gas constant (R_{air})	0.287	kJ/kg.K	$R_{air} = R_u/M$	A
	Atmospheric pressure	101.3	kPa		A
	Density	1.201	kg/m^3	ρ_{air} - Assume ideal gas properties (Pv = RT) (ρ = P/RT)	A
	Diesel fuel properties				
	Energy content	44	MJ/kg		A
	Density of Fuel	0.86	kg/L		A
	Conv_Density of Fuel	860	kg/m^3		A
	404D-22 Perkins 2.2L Diesel Engine				
	Maximum Power	38	kW		A
	Maximum Torque	142.7	Nm	@1800 rpm	A
	Number of Cylinder	4	ul		A
	Bore	84	mm		A
	Stroke	100	mm		A
	Displacement	2.2	L		A
	Conv_Displacement	0.002217	m^3	V_d - Volume displacement (Cyl swept volume x No cyl)	A
	Compression ratio	23.3	:1		A
	Revolutions per cycle	2	ul	n_r	A
	Manometer properties				
	Manometer fluid density	1000	kg/m^3	ρ_{man}	A
	Orifice diameter	0.03	m		A
	Orifice Area	0.000707	m^2	A_0	A
	Orifice correction factor	0.6	ul	C_0	A

Experimental Data

Area	Variable	Value						Units	Reference	Rev
	Engine parameters								Values are for each nominal engine speed	
	Nominal speed	1400	1400	1800	1800	2000	2000	rpm		A
	Engine load	50	100	50	100	50	100	%		A
	Measured speed	1390	1390	1793	1792	2000	2000	rpm		A
	Throttle position	68	100	85	100	96	100	%		A
	Power	8.7	17.7	10.6	22.1	11.6	22.9	kW		A
	Torque	59	122	55	118	56	109	Nm		A
	Temperatures									
	Water in temp.	21.9	21.26	22.9	22.3	24.1	23.4	°C	TC3	A
	Water out temp.	26.9	31.93	31.1	34.9	33.2	37.5	°C	TC4	A
	Cal. In temp.	174	302	202	360	214	367	°C	TC5	A
	Cal. Out temp	117	143	127	218	140	230	°C	TC6	A
	Exh. In temp	242.6	471.5	271.6	508.3	289.4	512.7	°C	TC1	A
	Exh. Out temp	218.6	408.0	251.5	451.6	267.5	461.8	°C	TC2	A
	Flow rates									
	Cooling water flow rate	51.5	40.4	41	40.4	40.4	40.4	L/min		A
	Fuel supply rate	0.054	0.098	0.059	0.118	0.062	0.124	L/min		A
	Air									
	Elevation difference (manometer)	122	112	202	194	255	230	mm		A

Calculation

Area	Variable	Value						Units	Reference	Rev
	Engine speed	1400	1400	1800	1800	2000	2000	rpm		
	Engine load	50	100	50	100	50	100	%		
	Mass flow rate of intake air									
	P_2-P_1	1.195	1.097	1.979	1.901	2.499	2.254	kPa	$P_2-P_1 = (\rho_{man} - \rho_{air})*g*(h_2-h_1)$	A
	Volume flow rate	0.0189	0.0181	0.0244	0.0239	0.0274	0.0260	m^3/s	$\dot{V}_{air} = C_0*A_0*\sqrt{2*(P_2-P_1)/\rho_{air}}$	A
	Conv_Volume flow rate	18.926	18.134	24.353	23.866	27.362	25.986	L/min		A
	Mass flow rate	0.0227	0.0218	0.0292	0.0287	0.0328	0.0312	kg/s		A
	Brake mean effective pressure									
	BMEP	0.334	0.692	0.312	0.669	0.317	0.618	MPa	$BMEP = 2*PI()*T*n_r/V_d$	A
	Conv_BMEP	3.345	6.916	3.118	6.689	3.175	6.179	Bar		A
	Break specific fuel consumption									
	Volume flow rate of fuel	9.00E-07	1.63E-06	9.83E-07	1.97E-06	1.03E-06	2.07E-06	m^3/s		A
	Mass flow rate of fuel	0.000774	0.001405	0.0008457	0.001691	0.000889	0.001777	kg/s		A
	Break power	8700.0	17700.0	10600.0	22100.0	11600.0	22900.0	W		A
	SFC	8.9E-08	7.94E-08	7.978E-08	7.65E-08	7.66E-08	7.76E-08	kg/J	$SFC = m_{fuel}/P_{brake}$	A
	Conv_SFC	320.3	285.7	287.2	275.5	275.8	279.4	g/kWh		A
	Average SFC			287.3				g/kWh		
	Air-Fuel ratio									
	Air-Fuel ratio	29.36	15.50	34.57	16.94	36.97	17.55	ul	$AFR = m_{air,in}/m_{fuel}$	A
	Volumetric efficiency									
	Actual air mass inducted per cycle	2.72E-05	2.61E-05	2.716E-05	2.66E-05	2.74E-05	2.6E-05	m^3/Rev.		A

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Volumetric efficiency	0.7371	0.7062	0.7353	0.7210	0.7406	0.7034	ul	$VE = n_r * m_{air,in} * 60/(\rho_{air} * V_d * N)$	A
Conv_Volumetric efficiency	73.71	70.62	73.53	72.10	74.06	70.34	%		A
Thermal efficiency									
Thermal efficiency	0.2555	0.2864	0.2849	0.2970	0.2967	0.2928	ul	$TE = P_{brake}/(m_{fuel} * LHV_{fuel})$	A
Conv_Thermal efficiency	25.55	28.64	28.49	29.70	29.67	29.28	%		A
Average Thermal efficiency			28.6				%		
Peak thermal efficiency			29.7				%		

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Energy balance calculation

Area	Variable	Value						Units	Reference		Rev
	Engine speed	1400	1400	1800	1800	2000	2000	rpm			
	Engine load	50	100	50	100	50	100	%			
	Heat transfer to cooling water										
	Denisty of water @ 25 degC	997	997	997	997	997	997	kg/m3			A
	Mass flow rate of cooling water (SI)	0.856	0.671	0.681	0.671	0.671	0.671	kg/s			A
	Cp @ 25 degC	4.18	4.18	4.18	4.18	4.18	4.18	kJ/kg.K	Assume to be constant		A
	Heat transfer	18.06	29.96	23.29	35.35	25.54	39.57	kW	Q = mcΔT		A
	Heat transfer through exhaust pipe										
	Mass flow rate of exhaust air	0.0235	0.0232	0.0301	0.0303	0.0337	0.0330	kg/s			A
	Cp air @ TC1	1.03294	1.08568	1.03892	1.09437	1.04276	1.09540	kJ/kg.K	Calculated using CoolProp + Python		A
	Cp air @ TC6	1.01301	1.01618	1.01416	1.02820	1.01579	1.03047	kJ/kg.K	Calculated using CoolProp + Python		A
	Cp,avg	1.02298	1.05093	1.02654	1.06129	1.02928	1.06294	kJ/kg.K			A
	Heat transfer	3.019	8.001	4.467	9.349	5.188	9.909	kW	Q = mcΔT		A
	Energies in										
	Chemical energy supplied by fuel	34.06	61.81	37.21	74.42	39.10	78.20	kW	Q_{fuel}		A
	Energies out										
	Shaft work	8.700	17.700	10.600	22.100	11.600	22.900	kW	W_{brake}	Measured break power output	A
	Heat removed by cooling water	18.064	29.955	23.292	35.346	25.535	39.566	kW	Q_{cool}	Calculated above	A
	Heat lost through walls of exh. Pipe	3.019	8.001	4.467	9.349	5.188	9.909	kW	$Q_{exh,pipe}$	Calculated above	A
	Unaccounted losses		6.149		7.624		5.828	kW	Q_{others}	$Q_{fuel} - (W_{brake} + Q_{cool} + Q_{exh,pipe})$	A